CAUSAFLOW AI

The CBM-Powered Parking Intelligence & Enforcement Engine

Final Submission Draft

One-line pitch: We don't just map where illegal parking happens. We estimate how much road capacity is lost, then help Bengaluru Traffic Police decide which zones to clear first.

1. Problem Statement and Theme Fit

The challenge is not simply to detect illegally parked vehicles. The theme asks for something more operational: identify illegal parking hotspots, quantify how much they disrupt traffic flow, and enable targeted enforcement. Most teams will stop at a heatmap or a raw violation counter. That looks useful, but it does not tell a traffic police officer where to act first, when to act, or why a zone deserves priority.

CausaFlow AI is built specifically to close that gap. It turns historical illegal parking records into a practical enforcement intelligence layer. Instead of saying only "this area has violations," it answers "this zone is consuming the most road capacity," "this is when the disruption is worst," and "this is the order in which enforcement resources should be deployed."

Theme alignment at a glance

Theme requirement	How CausaFlow AI delivers
Detect illegal parking hotspots	HDBSCAN spatio-temporal clustering on validated violation coordinates, anchored to junction names and recurring temporal patterns.
Quantify impact on traffic flow	Capacity-Blockage Minutes (CBM), an estimated lane-minute capacity loss metric based on duration, obstruction type, vehicle footprint, and junction sensitivity.
Enable targeted enforcement	A ranked zone list with Monitor / Patrol / Tow action tiers, time-window recommendations, and station-aware deployment guidance.

2. Core Idea

CausaFlow AI has one job: transform parking violations into enforcement-ready intelligence.

The system does three things in sequence:

- Detects persistent illegal parking hotspots from historical records.
- Estimates the traffic capacity lost by each violation and each zone.
- Ranks zones by operational priority so police can act where intervention matters most.

That is the whole design. Nothing is added just for show. Every module exists because it contributes to theme alignment, feasibility, or judge clarity.

3. Why this version is stronger than a heatmap

A basic heatmap answers only one question: where do violations happen? That is not enough for the challenge. A zone with fifty minor violations is not always worse than a zone with ten severe blockages near a critical junction. CausaFlow AI therefore adds a second layer: impact estimation.

The core differentiator is CBM, or Capacity-Blockage Minutes. It is a transparent, explainable metric that converts a parking violation into estimated lane-minute loss. That gives the police a practical ranking signal instead of a passive visual layer.

The important point is not that CBM is magical. It is that CBM is operational. A traffic officer can understand why a zone scored high, and a judge can follow the logic without needing a black-box model explanation.

4. Dataset Utilization

The solution is intentionally built around the provided CSV so that the prototype stays feasible within the hackathon window. Each meaningful column is used directly or indirectly.

CSV column	Usage in CausaFlow AI
latitude / longitude	Map matching and spatial clustering.
created_datetime	Time-of-day analysis and peak-hour patterns.
closed_datetime	Violation duration, which feeds CBM.
violation_type (JSON)	Obstruction type such as double-parking, bus-stop blocking, or footpath encroachment.
vehicle_type	Vehicle footprint weighting through a PCE-inspired multiplier.
junction_name	Hotspot anchoring and junction sensitivity.
police_station	Beat-aware enforcement guidance.
offence_code	Parking-specific filtering.
validation_status	Approved records used as ground truth; rejected records help identify ambiguity and noise.

5. How the model works

The architecture has three stages.

Stage 1: Hotspot detection

HDBSCAN groups recurring violations into persistent spatial-temporal clusters. That gives the system hotspots rather than isolated points.

A cluster becomes a hotspot when it has enough density, enough recurrence over time, and clear operational significance around roads, junctions, or commercial corridors.

Stage 2: Impact quantification

Each violation is converted into CBM, an estimated lane-minute capacity loss metric.

The simplified prototype formula is:

CBM = Duration × Lane-Blockage Factor × Vehicle Footprint × Junction Sensitivity

Duration is taken from closed_datetime minus created_datetime. If the duration is missing or unrealistic, the prototype caps it and imputes a median value by violation type and time bucket so the metric remains stable.

Lane-blockage factor reflects how severe the obstruction is. For example, double-parking is more disruptive than a light encroachment. Vehicle footprint uses a PCE-inspired weight so larger vehicles carry more impact. Junction sensitivity smooths the effect of proximity to critical junctions, so a blockage near a major junction receives more weight than one deep inside a low-pressure street segment.

Important wording for the final submission: CBM is an estimated capacity-loss metric, not a claim of sensor-level exact delay measurement. That keeps the system defensible in Q&A while still answering the theme directly.

Stage 3: Enforcement prioritization

Violations are aggregated by zone. The system then computes peak-hour intensity, recurrence score, and total CBM. From there it assigns action tiers:

- Top-impact zones: Tow priority
- Medium-impact zones: Patrol priority
- Low-impact zones: Monitor

The output is not a passive dashboard. It is a ranked deployment list with a recommended time window and a station-aware action plan.

6. The final formula and why it is defensible

CBM is deliberately simple enough to explain but rich enough to be useful. The multiplication structure ensures that a zone becomes truly important only when several factors are strong at the same time: long duration, severe obstruction, large vehicle footprint, and high junction sensitivity.

That matters because enforcement teams do not need mathematical decoration. They need a number they can trust, understand, and act on.

The model is also safer than a black-box prediction because every component can be explained in plain language. Judges can question the assumptions, and the team can answer them clearly.

7. What the dashboard will show

The prototype dashboard will surface the following outputs:

- An interactive Bengaluru map with clustered hotspots.
- A ranked enforcement table with zone name, CBM, action tier, and recommended patrol window.
- A zone drill-down view showing violation type mix, vehicle mix, time pattern, and recurrence.
- A simple rationale panel explaining why a zone received its ranking.

This is important for judge clarity. A judge should be able to see the logic in under one minute.

8. Feasibility and build plan

The concept is designed to be built quickly and without dependence on live cameras, third-party APIs, or fragile external systems. That makes it hackathon-safe.

Day	Build task
1	Clean data, parse JSON fields, compute duration, and generate CBM per record.
2	Run HDBSCAN clustering and form zones.
3	Aggregate CBM by zone and implement action-tier logic.
4	Build Streamlit dashboard with the map, ranked table, and drill-down view.
5	Pre-compute outputs into static JSON and create backup demo assets.
6	Rehearse the demo and prepare Q&A responses.

9. Risks and how the final draft handles them

Every serious hackathon idea has risk. The point is not to pretend the risks do not exist. The point is to show that they have already been thought through.

- Bias risk: police ticket data reflects enforcement patterns. CausaFlow explicitly acknowledges this and reduces the problem by combining recurrence, junction sensitivity, and time patterns instead of relying on ticket count alone.
- Duration noise: closed_datetime may be noisy or missing. The prototype caps outliers and imputes a sensible median when needed.
- Overclaiming risk: the draft never says CBM is exact sensor-grade delay. It says estimated capacity loss, which is safer and more honest.
- Demo risk: all outputs are pre-computed into static JSON, so the live demo is stable.
- Scope risk: the system stays focused on parking-induced congestion, not full city traffic simulation.

10. Q&A safety

The strongest version of the idea is not the loudest one. It is the version that can survive questions without collapsing. The final wording is intentionally careful.

If asked whether this is just a weighted score, the answer is that CBM is a capacity-based proxy built from traffic-engineering logic, not a random ranking formula. If asked about bias, the answer is that the system acknowledges enforcement bias and attempts to reduce it rather than deny it. If asked why HDBSCAN, the answer is that density-based clustering is the right fit for recurring spatial violations. If asked whether the system replaces human judgment, the answer is no: it supports human decision-making.

11. Final pitch for submission

CausaFlow AI converts historical parking violations into hotspot intelligence, estimated capacity-loss metrics, and enforcement priorities. It gives Bengaluru Traffic Police a simple answer to a hard question: where should we act first to recover the most road capacity?

12. Final evaluation

Criterion	Rating
Theme fit	9.5/10
Feasibility	10/10
Judge clarity	10/10
Q&A safety	9/10
Winning potential	High

Final locked position: this is the strongest version because it is focused, buildable, explainable, and directly aligned with the challenge theme. It does not try to impress with unnecessary complexity. It wins by answering the problem clearly and operationally.